TEXAS A&M UNIVERSITY Engineering

University of Cyprus

Hardware-in-the-Loop Laboratory Performance Verification of HVAC equipment

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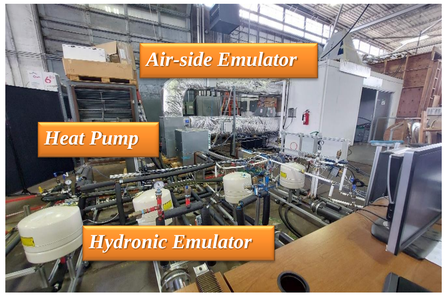
Introduction

HVAC Hardware-in-the-loop (HIL) Testbed: To study the operation dynamics of a Heating Ventilation and Air Conditioning (HVAC) system, a HIL testbed was used integrating simulation and experiment under climates of different cities in U.S.



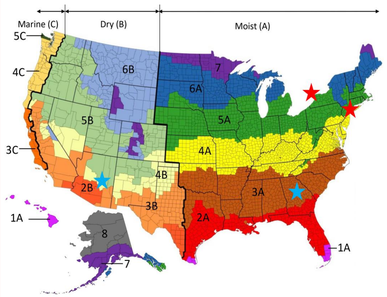
Heat Pump: For the construction of the test bed, different components such as heat exchangers, chillers, pipes, sensors, emulator coils were used, with the most important one to be the heat pump. Heat Pump is the element that provides either the heating or the cooling of the air, without using fossil fuels, since it uses electricity.

Objectives

Simulations and Testing: For the tests, four different climate conditions were used, using data from different cities (Tucson, Atlanta, New York, Buffalo). Moreover, different occupancy patterns were examined using LabVIEW. (72 Scenarios)

Objective:

1) Create a Manual for servicing the different elements
2) Use different statistical methods to compare the heat pumps results after the simulation, with theoretical simulation values.
3) Compare all data sets together to obtain the false variable, if one exist.



Methods

```
function [Power,Sim_Time,Tz_set,Tz,q,Tw,wz,Ret_Air_RH]=csv(Power,Sim_Time,Tz_set,Tz,q,Tw,wz,Ret_Air_RH,listing)
% Read CSV
for i=1:height(listing)
file_name=listing(i).name;
table=readtable(file_name);
% Parse Out Proper Data Columns
%HP_Mode=table.HPMode_1_11_; % Heat Pump Mode ([1,11])
%Fan_Speed=table.FanSpeedPercent___; % Fan Speed Percentage (%)
Power=vertcat(Power,table.WSHPPower_kW_); % Heat Pump Power (W)
Sim_Time=vertcat(Sim_Time,table.SimulationTimeStep); % Simulation Timestep
Ret_Air_RH=vertcat(Ret_Air_RH,table.WSHPReturnAirRH__RH_); % Relative Humidity
Tz_set=vertcat(Tz_set,table.SimCoolSP__C_); % Cooling Temperature Setpoint (Deg C)
Tz=vertcat(Tz,table.WSHPReturnAirTemp__C_); % Heat Pump Return Air Temp (Deg C)
q_sens=vertcat(q_sens,table.MeasSensibleload_kW_); % Heat Pump Sensible Capacity (kW)
q_lat=vertcat(q_lat,table.MeasLatentload_kW_); % Heat Pump Latent Capacity (kW)
q=vertcat(q,-q_sens+q_lat); % Heat Pump Total Capacity (kW)
Tw=vertcat(Tw,table.InletWaterTemp__C_); % Heat Pump Inlet Water Temp (Deg C)
wz=vertcat(wz,table.WSHPAvgReturnHR_kg_kg_); % Heat Pump Return Humidity Ratio (kg/kg)

sim_time_step_z=z.SimulationTimeStep;
ret_temp_z=z.WSHPAvgReturnAirTemp__C_;
ret_hum_z=z.WSHPAvgReturnHR_kg_kg_;
in_wat_z=z.InletWaterTemp__C_;
setpoin_z=z.SimCoolSP__C_;
WSHP_power_z=z.WSHPPower_kW_;
sim_zone_temp_z=z.SimZoneTemp__C_;
x=[sim_time_step_x, ret_temp_x, ret_hum_x, in_wat_x, setpoin_x, WSHP_power_x,sim_zone_temp_x ];
y=[sim_time_step_y, ret_temp_y, ret_hum_y, in_wat_y, setpoin_y, WSHP_power_y,sim_zone_temp_y];
z=[sim_time_step_z, ret_temp_z, ret_hum_z, in_wat_z, setpoin_z, WSHP_power_z,sim_zone_temp_z];
j=0; new_x=[];new_y=[]; new_z=[];
```

Part of the Code Used for Data Analysis

Different methods were used for the analysis of the data, with a combined code of about 1200 lines.

Methods:

1) Examine data of different occupancy modes
2) Examine data of different weather conditions
3) Analyze data using root mean square and percentage error
4) Variables: Return Air Temperature, Inlet Air Temperature, Humidity Ratio and Set point Temperature

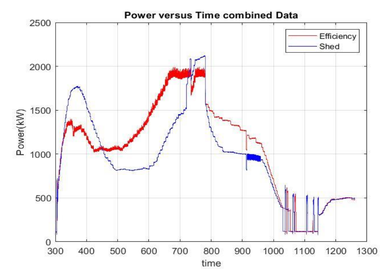
Keep three out of the four parameters constant and analyze the change in power consumption of the heat pump, to find if there is a wrong pattern

Results

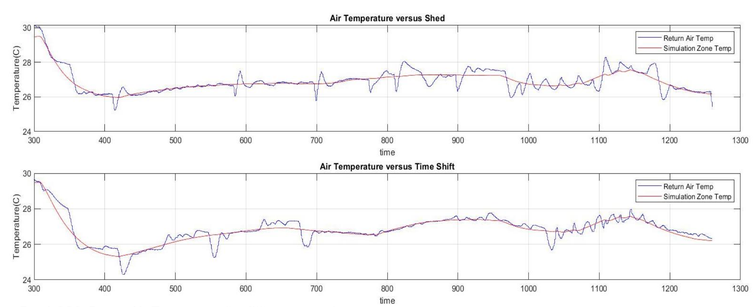
1) **Manual:** Service Manual for the different elements was created by combining the manuals of each element, also instructions for the correct use are mentioned as well.

2) **Data of different occupancy Modes:** Evaluate the differences of different test cases at the same location via the air temperature and power consumption to assist understanding of their accuracy. The graphs were plotted for each data to find a connection.

Note: Graphs and analysis was carried out for every data set using MATLAB.

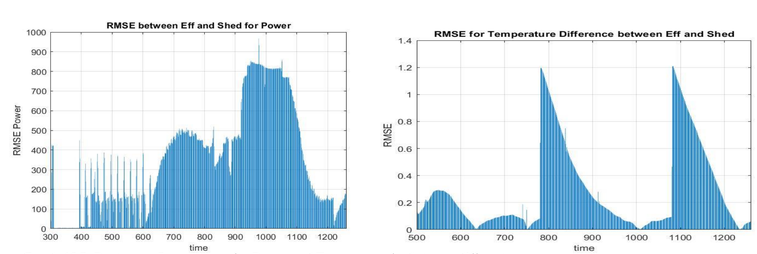


Graph 1. Power representation against time



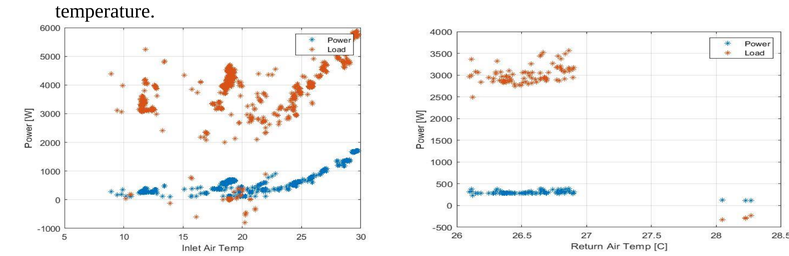
Graph 2&3. Return Air Temperature for different scenarios

3) **Use RMSE and percentage error:** Using statistical methods to understand the impact of the variation for different parameters on the system emulation accuracy (RMSE)



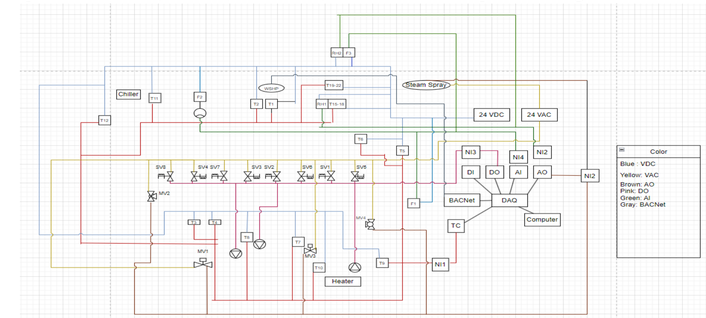
Graph 4&5. Root mean Square error for Power and Temperature for between different cases

4) **Combine all data sets:** Analyze the distribution of power and heating/cooling load for all simulations and detect the outliers in terms of inlet air temperature and return air temperature.



Graph. 6&7. Power Graph against the parameter that was kept constant

5) **Test bed's Electrical Schematic:** Generated detailed schematic of electrical connections for powered devices, sensors, controllers, and actuators for operation diagnosis and maintenance.



Test Bed Electrical Schematic

Conclusions

1) **Conclusion from 2nd part of the Results:**

- Cities that have hotter climates have major differences in the power graphs
- Big fluctuations for a small-time changes significantly the whole graph
- The inability of the real model to keep exact track of the simulations, seems to be a major issue, of the power's graph shape

2) **Conclusions from the 3rd part of the results:**

➢ **Comparison of total Load and Power**

- The RMSE values, have a constant relationship
- Some fluctuations were observed
- The constant relation might be affected from different factors as well
- No need for more locations to be analyzed
- No further test are needed at this set of data, after the results

➢ **Compare the Temperature difference with the power of the WSHP and different set point temperatures**

- Efficiency and shedding RMSE graphs are similar with efficiency and shifting RMSE
- Only efficiency case - has shown a possibility, that the occupancy are not as significant as the different set point cases

3) **Conclusion of the 4th part of the Results**

- Load data are scrambled and does not give a clear understanding
- Humidity Ratio seems to have no significant impact on the power consumption
- The increase in Inlet Water Temperature seems to have a logarithmic behavior
- Constant power values were observed at high set point temperatures
- Return Air Temperature seems to be around a mean value

Acknowledgments

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